

Patient Management System

Full Technical Documentation

UoVT SOF 23/24 Semester 3

Visual Programming II Group Assignment

- **SOF/23/B2/29** - Nuwan Hasanka
- **SOF/23/B2/02** - Ramesh Madara
- **SOF/23/B2/23** - Gihan Lavnidu
- **SOF/23/B2/28** - Chiranthi Bhagya
- **SOF/23/B2/20** - Suresh Indika
- **SOF/23/B2/14** - Thisari Wijerathne

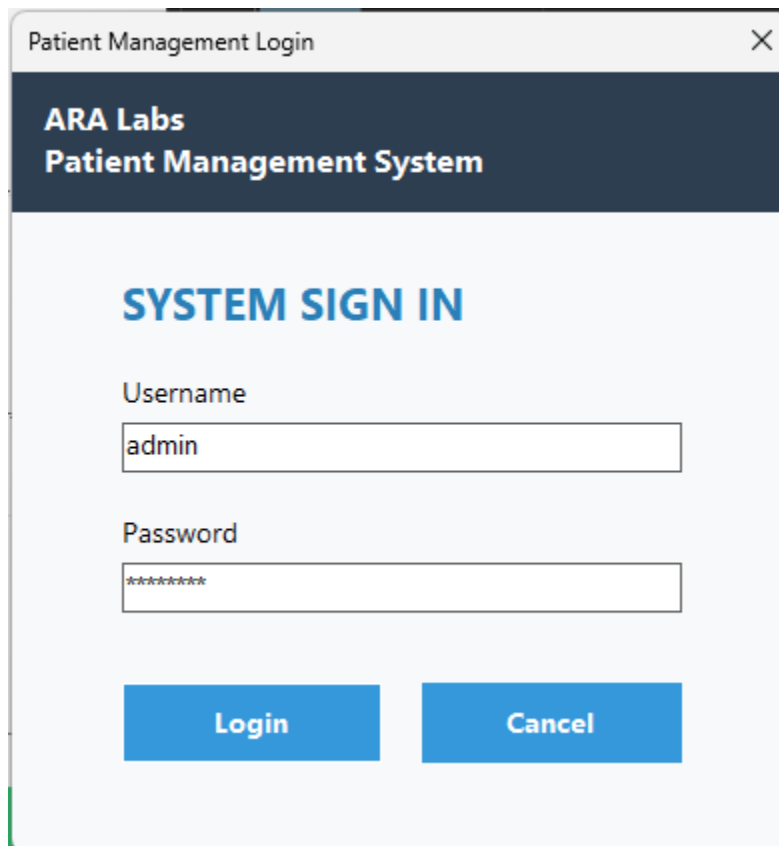
Overview

A secure, desktop-based Windows Forms application built with C# and .NET for managing patient clinical records. This system focuses on security, usability, and professional data management for medical clinics.

Features & User Flow

1. Launch & Secure Login

The user starts the application and logs in securely. The system features differentiated access levels for **Admins** and **Doctors**. Passwords are encrypted for safety.



The screenshot shows a web application window titled "Patient Management Login" with a close button (X) in the top right corner. The header area is dark blue with the text "ARA Labs" and "Patient Management System" in white. Below the header, the main content area is light gray and features the heading "SYSTEM SIGN IN" in blue. There are two input fields: "Username" with the text "admin" and "Password" with masked characters "*****". At the bottom, there are two blue buttons labeled "Login" and "Cancel".

2. Patients Dashboard

Once logged in, staff are greeted by the main dashboard. The **Patients Dashboard** allows for real-time search (by Name or NIC), sorting, and immediate access to patient profiles and histories.

ARA Labs
Patient Management System

Patients

View History

Staff Management

Manage Templates

Add Record

Logout

PATIENTS DASHBOARD

Search Name or NIC...

All Genders

+ Patient

ID	Name	NIC	Age	Contact	Gender	Edit	Delete
1	Nuwan	199324100400	30	0750891059	Male		
2	Sahan Hasanka	199324398300	30	0750894056	Male		
3	Madara	982040639v	30	1233211234	Male		
4	Sandamali	123456789V	12	1234567890	Female		

User: admin | Role: Admin

3. Encrypted Medical Records

Doctors can securely add diagnoses and prescriptions.

High Security: All clinical data (Diagnosis, Prescription, and dynamic test results) is encrypted using AES-256 *before* being stored in the SQL Database to ensure maximum patient privacy.

ARA Labs
Patient Management System

Patients

Add Record

View History

Manage Templates

Staff Management

Logout

SECURE MEDICAL RECORD Entry

Search by Name/NIC * Nuwan - 199324100400

Search

Selected Patient: Nuwan | Gender: Male | Blood: O+

Prescription

Medical Report

Diagnosis (AES Encrypted)

Flu

Prescription Details (AES Encrypted)

Flu Meds

Save Encrypted

Clear

User: doctor | Role: Doctor

4. Dynamic Medical Reports

Admins can build custom report templates (e.g., Blood Tests, RT-PCR) by defining specific fields. Doctors can then select a template, and the UI will automatically generate the corresponding input fields on-the-fly. All report data is stored using an Entity-Attribute-Value (EAV) model and encrypted.

The screenshot displays the 'Secure Medical Clinic System - Dashboard' interface. On the left is a dark sidebar with the following menu items: 'ARA Labs Patient Management System' (with a logo), 'Patients', '+ Add Record' (highlighted in blue), 'View History', 'Manage Templates', 'Staff Management', and a red 'Logout' button at the bottom. The main content area is titled 'SECURE MEDICAL RECORD Entry'. It features a search bar with the text 'Search by Name/NIC *' and a dropdown menu showing 'Sandamali - 123456789V', with a blue 'Search' button. Below the search bar, it states 'Selected Patient: Sandamali | Gender: Female | Blood: A+'. There are two tabs: 'Prescription' and 'Medical Report' (which is active). The 'Medical Report' section is titled 'DYNAMIC MEDICAL REPORT: Sandamali' and includes a 'Select Template:' dropdown menu set to 'COVID-19 RT-PCR'. Below this is a form with a 'SARS-CoV-2 Result' dropdown menu set to 'Positive', a 'Physician Notes' section with a text input field containing 'Totally Positive :(', and a blue 'Save Report' button. At the bottom of the dashboard, a status bar shows 'User: doctor | Role: Doctor'.

5. Professional PDF Export

Staff can click "View Record" to read decrypted details in a beautifully designed popup, and then click **Export PDF** to generate an official, printable document. The generated PDFs feature automated formatting, highlighting, timestamps, and signature lines in a signature sticky-note aesthetic.

PDF Preview

ARA Labs

Air Port Junction Ratmalana, Colombo
Tel: 011-2345678 | Web: www.aralabs.lk

OFFICIAL REPORT: COVID-19 RT-PCR

Patient Name: Sandamali

NIC Number: 123456789V

Contact: 1234567890

Address: test

Report ID: 11

Date: Monday, September 28, 2026 11:34 PM

Gender: Female

Age: 12

Blood: A-

TEST RESULTS

Parameter	Result	Ref Range	Flag
SARS-CoV-2 Result	Positive *	Negative	ABNORMAL
Physician Notes	Totally Possitive :(		

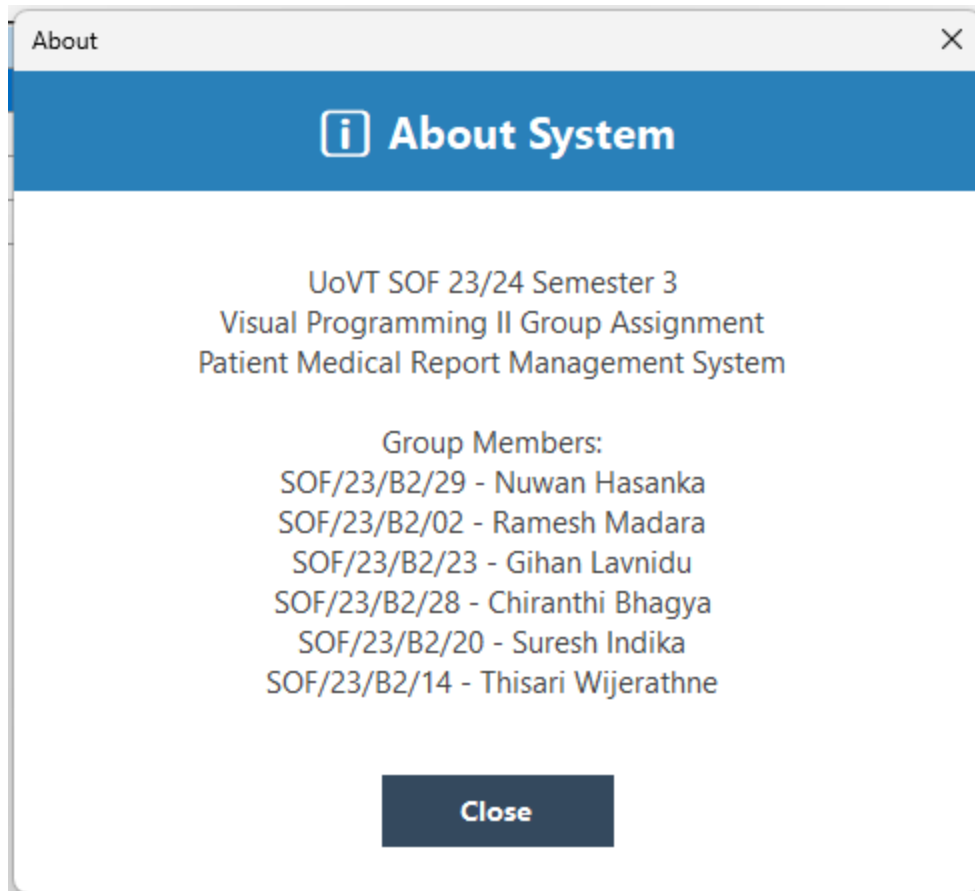
Authorized Signature: _____

Save PDF

Close

6. System Information & UI Polish

The application was built with a strict adherence to a modern flat UI design, doing away with outdated 3D borders for a crisp, responsive, and professional user experience.



Design Guidelines

Visual Language

The application adopts a **Modern Flat UI Design** built entirely in Windows Forms, departing from the classic 3D gray UI in favor of solid colors, crisp typography, and borderless elements.

Color Palette

The color scheme is inspired by professional medical and enterprise dashboards, heavily relying on the Flat UI color palette:

- **Sidebar Background:** #2c3e50 (Midnight Blue) - Grounding the navigation.
- **Main Background:** #ffffff (White) - Keeping the data area clean and legible.

- **Primary Action / Active Tabs:** #2980b9 (Belize Hole / Bright Elegant Blue) - Used for primary actions, headers, and active tab highlights.
- **Success / Add:** #2ecc71 (Emerald) - Used for creating patients and saving records.
- **Destructive / Logout:** #c0392b / #e74c3c (Alizarin) - Used for logout and deletion operations.
- **Inactive / Disabled:** #f0f0f0 (Light Gray) - Used for inactive tabs or disabled fields.
- **Text (Primary):** #404040 (Dark Gray) for readability over pure black.
- **PDF Documents:** #ffffcc (Light Yellow) - Used for the signature "Sticky Note" aesthetic when generating PDF exports.

Component Styling

- **Buttons:** FlatStyle.Flat with BorderSize = 0. White foreground text.
- **Data Grids:** EnableHeadersVisualStyles = false, customized header backgrounds to match the primary theme, no cell borders, alternating row colors for readability.
- **Tabs:** OwnerDrawFixed custom rendering. Active tabs are highlighted in bright blue.

Database Architecture

Server: .\SQLEXPRESS

Database: medicaldb

Schema Overview

- **Users:** Stores authentication and role information for system access.
- **Patients:** Stores core demographic data (Name, NIC, Contact, Address, BloodGroup, Gender, Age).
- **MedicalRecords:** Stores standard prescription and diagnosis records for patients. (Encrypted at rest).
- **ReportTemplates:** Stores dynamic template structures for specialized medical reports.
- **TemplateFields:** Stores the individual fields belonging to a specific ReportTemplate.
- **PatientReports:** Stores the metadata linking a patient to a generated dynamic report.

- **ReportResults:** Stores the actual values entered into the fields of a dynamic report. (Encrypted at rest).

Technology Stack

- **Frontend:** C# Windows Forms (WinForms)
- **Backend:** .NET Framework 4.8
- **Database:** Microsoft SQL Server Express
- **Data Access:** ADO.NET (System.Data.SqlClient)
- **Cryptography:** System.Security.Cryptography.Aes for AES-256 data encryption.
- **Reporting:** iTextSharp (v5.5.13.3) for generating stylized PDF documents.
- **Iconography:** Unicode Emoji Icons to maintain a lightweight footprint without external assets.

Technical Walkthrough

UI/UX Engine

UITheme.cs: A centralized static class that recursively traverses any form passed to it. It flattens buttons, styles grids, removes borders, and enforces the global font (Segoe UI) and color palette, ensuring absolute consistency across the application.

Encryption Pipeline

Data strings are fed through `Aes.Create()`, written to a `CryptoStream`, and converted to Base64 strings for storage. When a doctor views a patient's history, the data is transparently decrypted in memory.

Dynamic Rendering Logic (EAV Model)

In `AddMedicalReportForm.cs`, the app queries `TemplateFields` for the selected template. A loop dynamically instantiates Label and Control (TextBox, NumericUpDown) pairs. On save, the system iterates over the dynamic

controls, extracts their values, encrypts them, and saves them to the database.